

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION



MATHEMATICS
3 UNIT (ADDITIONAL)
AND
3/4 UNIT (COMMON)

DIRECTIONS TO CANDIDATES

Name

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed on the last page.
- Board-approved calculators may be used.
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

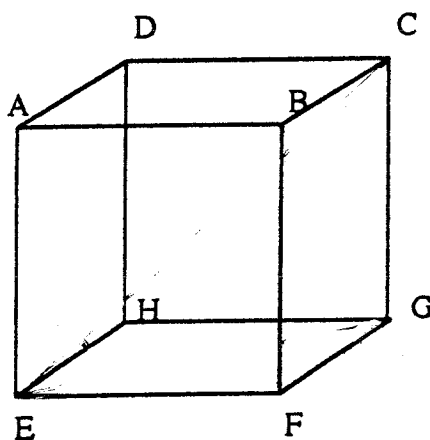
This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 1996 HSC examination paper in this subject.

QUESTION 1 (start a new page)

- a) Solve $\frac{2x+5}{x+1} < 1$
- b) Find the co ordinates of the point that divides the interval joining A(-1, 4) to B(7, 12) externally in the ratio 1 : 2
- c) Differentiate with respect to x ;
- i) $y = \sqrt{\sin x}$
- ii) $y = \sin^{-1}(1 - x)$
- d) Evaluate $\int_{0.1}^{0.4} \sec^2 3x \, dx$ correct to 3dp

QUESTION 2 (start a new page)

- a) Find the exact value of $\sin^{-1}\left(\frac{1}{2}\right) - \tan^{-1}(-\sqrt{3})$
- b) For the polynomial $P(x) = x^3 + x - 1$
- i) show that a root exists between $x = 0$ and $x = 1$
- ii) use one approximation of Newtons' method to achieve a better estimate of this root which lies near 0.5 correct to 2 decimal places.
- c) Find the equation of the tangent to $y = \tan 3x$ at the point where $x = \frac{\pi}{3}$
- d) The figure below is a cube.



Calculate the angle between CE and the plane EFGH (answer to the nearest minute).

QUESTION 3 (start a new page)

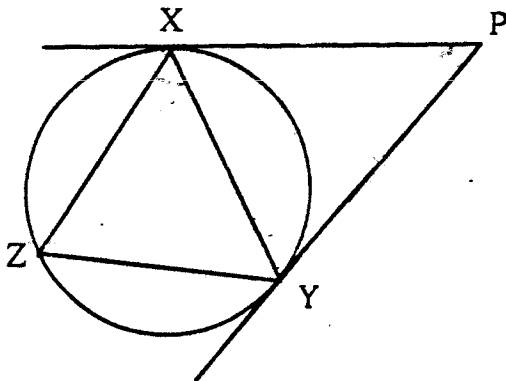
- a) Given the function $y = 3\sin(2x + \pi)$;
i) state the period and amplitude
ii) sketch the graph for $0 \leq x \leq 2\pi$
- b) Using the substitution $u = 1 + x^2$ find $\int x(1 + x^2)^7 dx$
- c) Use mathematical induction to show that $3^{2^n} - 1$ is divisible by 8

QUESTION 4 (start a new page)

- a) i) For what value of k is the polynomial $Q(x) = 4x^3 - x + k$ divisible by $2x + 3$?
ii) Use your answer from i) to fully factorise $Q(x)$
- b) P is a point on the parabola $x^2 = 4y$. Show the normal to the curve at $P(2p, p^2)$ has equation $x = -py + 2p + p^3$
- c) Given the function $6\cos^2 \theta + 8\sin \theta \cos \theta$
i) Express the function in terms of $\cos 2\theta$ and $\sin 2\theta$
ii) Hence deduce an expression for the function in the form $A + 5\cos(2\theta - \alpha)$ where A and α are constants.
iii) Solve the equation $6\cos^2 \theta + 8\sin \theta \cos \theta = 4$ for $0^\circ \leq \theta \leq 360^\circ$

QUESTION 5 (start a new page)

- a) i) Show that $P = P_0 e^{kt}$ satisfies the equation $\frac{dP}{dt} = kP$
- ii) In a culture of bacteria the number present P , is given by the formula $P = P_0 e^{kt}$ where P_0 is the initial population of bacteria and k is a constant. If between 1am and 4am the population doubles, at what time would you expect the population to be ten times the 1am population?
- b) The velocity of a particle is given by $v = 2x + 1 \text{ cms}^{-1}$. If the initial displacement is 1 cm to the right of the origin, find the displacement as a function of time.
- c)

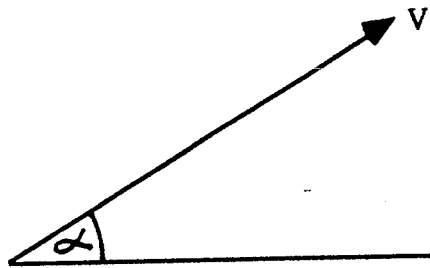


PX and PY are tangents, $\angle YXZ = \angle XPY = 2a^\circ$, prove $XZ = XY$ giving reasons.

QUESTION 6 (start a new page)

a) Find the volume of the solid of revolution generated by rotating $y = \sin x$ around the X axis from $x = 0$ to $x = \pi$

b)

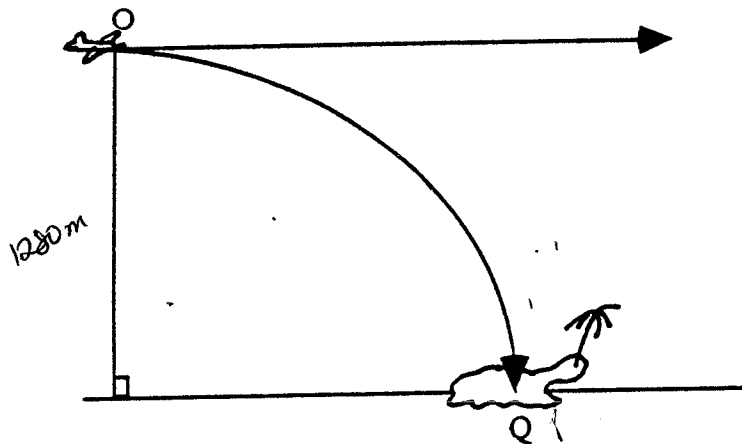


i) A particle is projected with a velocity $V \text{ ms}^{-1}$ at an angle α to the horizontal. Show that the projectile's trajectory is defined by the equations

$$x = Vt \cos \alpha$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \alpha$$

ii) A plane is flying horizontally at 400 ms^{-1} at a height of 1280m above the ocean. It releases a survival package from a point O towards the centre of a small island Q



α) How far before Q should the package be released so that it falls on the centre of the island? (use $g = 10 \text{ ms}^{-2}$)

β) Show that the speed of the package on impact is approximately 431 ms^{-1}

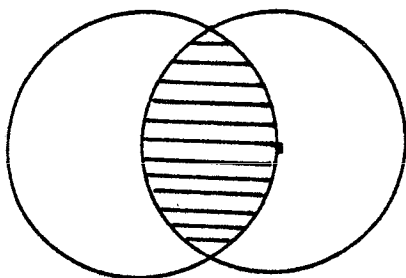
QUESTION 7 (start a new page)

a) i) Show that $\frac{5}{(x-2)(x+3)}$ can be expressed in the form $\frac{A}{x-2} + \frac{B}{x+3}$

ii) Hence or otherwise find $\int \frac{5 dx}{(x-2)(x+3)}$

b) On a certain day in Fremantle Harbour the depth of high tide is 32 metres. At low tide $6\frac{1}{2}$ hrs later the depth of water is 21 metres. If ^{low} high tide is 12.10am, what is the earliest time at which a ship needing 28.5 metres of water can enter the harbour. (Assume rise and fall of tide in SHM)

c)



Two equal circles of radius r are drawn passing through the centre of each other.

Show that the common area is $r^2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$ units²

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Stephane SIM.

8/7/02

Sydney Girls High School 1996 3/4 U.

QUESTION 1

A.) $\frac{2x+5}{x+1} - 1 < 0$

$\frac{2x+5-x-1}{x+1} < 0 \quad ; \quad \frac{x+4}{x+1} < 0$

① $x+4 < 0$
 $x < -4$

$x+1 > 0$
 $x > -1$

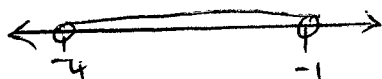


✓

∴ no solution

② $x+4 > 0$
 $x > -4$

$x+1 < 0$
 $x < -1$



∴ $-4 < x < -1$ ✓

B) let point be $P(p, q)$

$P = \left(\frac{1(7) - 2(-1)}{1-2}, \frac{1(12) - 2(4)}{1-2} \right)$

$= \left(\frac{7+2}{-1}, \frac{12-8}{-1} \right)$

$= (-9, -4)$ ✓

$$\begin{aligned} \text{c) i.) } y &= \sqrt{\sin x} = (\sin x)^{\frac{1}{2}} \\ \frac{dy}{dx} &= \frac{1}{2} (\sin x)^{-\frac{1}{2}} \times \cos x \\ &= \frac{\cos x}{2\sqrt{\sin x}} \times \frac{\sqrt{\sin x}}{\sqrt{\sin x}} = \frac{\sqrt{\sin x} \cos x}{2 \sin x} = \frac{\sqrt{\sin x} \cot x}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{ii.) } y &= \sin^{-1}(1-x) \\ \frac{dy}{dx} &= \frac{1}{\sqrt{1-(1-x)^2}} \times -1 \\ &= \frac{-1}{\sqrt{1-(1-2x+x^2)}} = \frac{-1}{\sqrt{2x-x^2}} \checkmark \end{aligned}$$

$$\text{D) } \int_{0.1}^{0.4} \sec^2 3x \, dx \quad \text{RADIAN!}$$

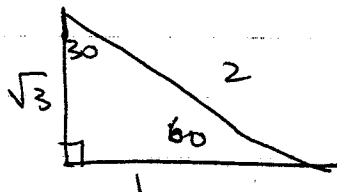
$$= \frac{1}{3} (\tan 3x) \Big|_{0.1}^{0.4} \checkmark$$

$$\begin{aligned} &= \frac{1}{3} \{ \tan 1.2 - \tan 0.3 \} \\ &= \frac{1}{3} 2.263 \text{ (3 dp)} = 0.754 \text{ rad (to 3 dp).} \end{aligned}$$

QUESTION 2

$$\begin{aligned} \text{A.) } \sin^{-1}\left(\frac{1}{2}\right) - \tan^{-1}(-\sqrt{3}) \\ = \sin^{-1}\left(\frac{1}{2}\right) + \tan^{-1}(\sqrt{3}) \checkmark \end{aligned}$$

$$= \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi + 2\pi}{6} = \boxed{\frac{\pi}{2}} \checkmark$$



B) $P(x) = x^3 + x - 1$

i.) $P(0) < 0$

$P(1) > 0$

$P(x)$ is a continuous function
 $\therefore$ a root exists b/w $x=0$ and $x=1$ ✓

ii.) $x_1 = 0.5$

$x_2 = x_1 - \frac{P(x_1)}{P'(x_1)}$ ✓

$P'(x) = 3x^2 + 1$

$x_2 = 0.5 - \frac{P(0.5)}{P'(0.5)}$

$= \underline{\underline{0.71}}$ (2dp) ✓

c) $y = \tan 3x$.

when $x = \frac{\pi}{3}$, $y = \tan \pi = \underline{0}$ ✓

$\therefore (\frac{\pi}{3}, 0)$ lies on the curve $y = \tan 3x$.

$\frac{dy}{dx} = 3 \sec^2 3x$ ✓

when $x = \frac{\pi}{3}$, $\frac{dy}{dx} = 3 \sec^2 \pi = \underline{3}$ ✓

Eqn of tgt is $(y-0) = 3(x - \frac{\pi}{3})$

$y - 3x + \pi = 0$
 $3x - y - \pi = 0$ ✓

- 4 -

1) let all sides of cube = x .

In $\triangle EFG$,

$$(EG)^2 = (EF)^2 + (FG)^2$$

$$(EG)^2 = x^2 + x^2$$

$$(EG)^2 = 2x^2$$

$$\underline{EG = \sqrt{2}x} \quad \checkmark \quad (EG > 0 \text{ because it is a length})$$

$$\text{In } \triangle ECG, \quad \tan \angle CEG = \frac{CG}{EG}$$

$$= \frac{x}{\sqrt{2}x} \quad \checkmark$$

$$\tan \angle CEG = \frac{1}{\sqrt{2}} \quad \frac{S}{\sqrt{T}} \bigg| \frac{A}{C} \checkmark$$

$$\angle CEG = \underline{35^\circ 16'} \quad \text{OR} \quad \underline{215^\circ 16'}$$

But angle is acute as shown in diagram.

$$\therefore \angle CEG = \underline{35^\circ 16'}$$

QUESTION 3

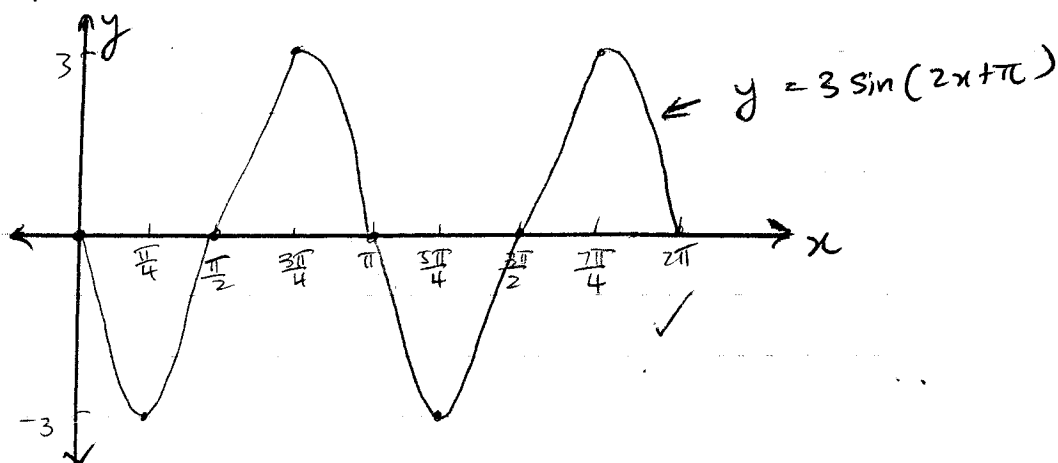
A)

$$y = 3 \sin(2x + \pi)$$

$$\therefore \text{Period} = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi \quad \checkmark$$

$$\text{Amplitude} = 3 \quad \checkmark$$

ii.)



B)

$$\int x(1+x^2)^7 dx$$

$$u = 1+x^2 \\ \frac{du}{dx} = 2x \quad ; \quad du = 2x dx$$

$$= \int \frac{1}{2} u^7 du \quad \checkmark$$

$$= \frac{1}{2} \int u^7 du = \frac{1}{2} \left(\frac{u^8}{8} \right) + C$$

$$= \frac{u^8}{16} + C = \frac{(1+x^2)^8}{16} + C \quad \checkmark$$

c) Step 1

Let $n=1$.

$$= 8(1)$$

$$3^{2n} - 1 = 3^2 - 1 = 8 \text{ which is divisible by } 8$$

$\therefore$ true for $n=1$ ✓

Step 2

Assume true for $n=k$

$$3^{2k} - 1 = 8M \text{ (M is an integer)}$$

R.T.P. also true for $n=k+1$

$$3^{2k+2} - 1 = 8N \text{ (N is another integer)}$$

$$\begin{aligned} \text{LHS } 3^{2k+2} - 1 &= 3^2(3^{2k} - 1) + 8 \quad \checkmark \\ &= 9(8M) + 8 = 8(9M + 1) \quad \checkmark \\ &= 8N = \text{RHS} \quad \checkmark \end{aligned}$$

Step 3

as well as
If $n=k$ is true and $n=k+1$ is true, ~~and~~ $n=1$ is also true,
then $n=1+1=2$ is true, $n=2+1=3$ is true and so on. $\therefore$
by TPOM1, it is true for all positive integers $n \geq 1$.

QUESTION 4

A) i.) For $Q(x)$ to be divisible by $2x+3$,

$$Q(-1.5) = 0.$$

$$Q(-1.5) = 4\left(-\frac{3}{2}\right)^3 + \frac{3}{2} + k = 0$$

$$4\left(-\frac{27}{8}\right) + \frac{3}{2} + k = 0 \quad \checkmark$$

$$k = -\frac{3}{2} + \frac{27}{2} \quad ; \quad k = \frac{24}{2} = 12$$

$$\therefore k = 12$$

$$\begin{array}{r}
 \text{ii.) } 2x+3 \sqrt{2x^2-3x+4} \\
 \underline{4x^3+0x^2-x+12} \\
 4x^3+6x^2 \\
 \underline{-6x^2-x} \\
 -6x^2-9x \\
 \underline{8x+12} \\
 8x+12 \\
 \underline{00} \\
 \therefore \underline{\underline{Q(x) = (2x+3)(2x^2-3x+4)}}
 \end{array}$$

B) $x^2 = 4y$

$$y = \frac{x^2}{4} ; \frac{dy}{dx} = \frac{x}{2}$$

Grad. of tgt at P = $\frac{2p}{2} = p$ ✓

∴ grad. of normal at P = $-\frac{1}{p}$ ($m_1 m_2 = -1$ for $\perp$ lines)

Eqn of normal at P is =

$$(y-p^2) = -\frac{1}{p}(x-2p)$$

$$p(y-p^2) + x - 2p = 0$$

$$py - p^3 + x - 2p = 0 ; \underline{x = -py + 2p + p^3}$$

c) $6\cos^2\theta + 8\cos\theta\sin\theta$

$$\begin{aligned}
 \text{i.) } & 6\cos^2\theta + 8\cos\theta\sin\theta \\
 &= 3(2\cos^2\theta - 1) + 3 + 4\sin 2\theta \\
 &= \underline{3\cos 2\theta + 4\sin 2\theta + 3} \quad \checkmark
 \end{aligned}$$

$$\cos 2\theta = 2\cos^2\theta - 1$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

ii.) $[3\cos 2\theta + 4\sin 2\theta] + 3 = R\cos(2\theta - \alpha) + 3$

$$= R(\cos 2\theta \cos \alpha + \sin 2\theta \sin \alpha) + 3$$

$$3 = R\cos \alpha \quad \text{--- (1)}$$

$$4 = R\sin \alpha \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} = \frac{R\sin \alpha}{R\cos \alpha} = \tan \alpha = \frac{4}{3}$$

$$\alpha = \underline{53^\circ 8'} \quad \checkmark$$

$$R = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = \underline{(5)} \quad \checkmark$$

$$\therefore 3\cos 2\theta + 4\sin 2\theta + \cancel{3} = 5\cos(2\theta - 53^\circ 8') + \cancel{3} \quad \checkmark$$

$$= 5\cos(2\theta - 53^\circ 8') + A$$

$$\therefore 3\cos 2\theta + 4\sin 2\theta = 5\cos(2\theta - 53^\circ 8')$$

$$\text{iii.) } 6\cos 2\theta + 8\sin \theta \cos \theta = 5\cos(2\theta - 53^\circ 8')^3 = 4 \quad 0^\circ \leq \theta \leq 360^\circ$$

$$\cos(2\theta - 53^\circ 8') = \frac{4}{5} \quad \frac{5}{1} \quad \checkmark \quad 0 - 53^\circ 8' \leq 2\theta - 53^\circ 8' \leq 720^\circ - 53^\circ$$

$$2\theta - 53^\circ 8' = -36^\circ 52', 36^\circ 52', 323^\circ 8', 396^\circ 52',$$

$$683^\circ 8', 75^\circ 28', 281^\circ 52', 438^\circ 27', 641^\circ 32' \quad \checkmark$$

$$2\theta = 16^\circ 16', 90^\circ, 376^\circ 16', 450^\circ$$

$$\boxed{\theta = 8^\circ 8', 45^\circ, 188^\circ 8', 225^\circ}$$

$$2\theta = 131^\circ 36', 334^\circ 40', 491^\circ 35'$$

$$\theta = 65^\circ 48', 167^\circ 20', 245^\circ 47', 347^\circ 20'.$$

QUESTIONS

$$\text{A.) i.) } P = P_0 e^{kt}$$

$$\frac{dP}{dt} = k P_0 e^{kt} = kP \quad \checkmark$$

$$\text{ii.) When } t=0, P = P_0 \quad \therefore P_0 \text{ is initial population.} \quad \checkmark$$

$$\text{When } t=3, P = 2P_0.$$

$$2P_0 = P_0 e^{3k} \quad ; \quad e^{3k} = 2$$

$$3k = \ln 2$$

$$k = \frac{\ln 2}{3} \quad \checkmark$$

$$\therefore P = P_0 e^{\frac{\ln 2}{3} t}$$

$$10P_0 = P_0 e^{\frac{\ln 2}{3} t} \quad \checkmark$$

$$e^{\frac{\ln 2}{3} t} = 10 \quad ; \quad \frac{\ln 2}{3} t = \ln 10$$

$$t = \frac{\ln 10 \times 3}{\ln 2} = 9 \text{ hrs } 58 \text{ min}$$

$$\therefore \boxed{\text{at } 10.58 \text{ am}}$$

B)

$$v = (2x+1) \text{ cm/s}$$

$$\frac{dx}{dt} = 2x+1$$

$$\frac{dt}{dx} = \frac{1}{2x+1}$$

$$t = \frac{1}{2} \ln(2x+1) + C$$

when $t=0$, $x=1$

$$0 = \frac{1}{2} \ln 3 + C; \quad C = -\frac{1}{2} \ln 3 = -\ln \sqrt{3}.$$

$$t = \frac{1}{2} \ln(2x+1) - \ln \sqrt{3} \quad \checkmark$$

$$t + \ln \sqrt{3} = \frac{1}{2} \ln(2x+1)$$

$$2t + 2 \ln \sqrt{3} = \ln(2x+1)$$

$$2x+1 = e^{2t+2 \ln \sqrt{3}}$$

$$x = \frac{e^{2t+2 \ln \sqrt{3}} - 1}{2} \quad \checkmark$$

$$x = \frac{e^{2t} \cdot e^{\ln 3} - 1}{2}$$

$$\boxed{x = \frac{3e^{2t} - 1}{2}} \quad \checkmark$$

c) $PX = PY$ (Tangents ^{drawn} from a pt outside a circle ~~drawn~~ are equal)

$\therefore \triangle PXY$ is an isos $\triangle$ (2 sides equal) $\checkmark$

$\therefore \angle PXY = \angle PYX$ (base $\angle$ of isos $\triangle$ are equal)

Let $\angle PXY = x$

$\angle PXY = \angle XZY = x$ (angle in alt. segment theorem).

$\angle YXZ = \angle XPY = 2a$ (given) $\checkmark$

In $\triangle PXY$, $2a + x + x = 180^\circ$ ($\angle$ sum $\triangle = 180^\circ$)

In $\triangle XZY$, $2a + x + \angle XYZ = 180^\circ$ (" ") $\checkmark$

$\angle XYZ = x$

$$\therefore \angle XYZ = \angle XYZ = x$$

$\therefore XZ = XY$ (ΔXYZ is an isos Δ . base $\angle$ equal.
2 sides also equal) ✓

QUESTION 6

$$\begin{aligned} \text{A)} \quad V &= \pi \int_0^\pi \sin^2 x \, dx \\ &= \frac{\pi}{2} \int_0^\pi (1 - \cos 2x) \, dx \quad \checkmark \\ &= \frac{\pi}{2} \left(x - \frac{1}{2} \sin 2x \right)_0^\pi \\ &= \frac{\pi}{2} \left(\pi - \frac{\sin 2\pi}{2} \right) \quad \checkmark \end{aligned}$$

$$\boxed{= \left(\frac{\pi^2}{2} \right) \text{ m}^3} \quad \checkmark$$

$$\begin{aligned} \text{B)}; \text{ i)} \quad \ddot{x} &= 0 \\ \dot{x} &= v \cos \alpha \\ x &= vt \cos \alpha. \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad x &= vt \cos \alpha \\ x &= 400t \cos 0 \\ \underline{x} &= 400t \end{aligned}$$

$$\begin{aligned} \ddot{y} &= -g \\ \dot{y} &= -gt + v \sin \alpha \\ y &= -\frac{gt^2}{2} + vt \sin \alpha. \quad \checkmark \end{aligned}$$

$$y = -\frac{10t^2}{2} + 400t \sin 0 + 1280$$

$$\underline{y = -5t^2 + 1280}$$

d) Find t when $y=0$

$$1280 - 5t^2 = 0 \quad ; \quad 5t^2 = 1280$$

$$t^2 = 256 \quad \checkmark$$

$$t = 16 \text{ s } (t > 0)$$

$$x = 400(16) = 6400 \text{ m} = \underline{\underline{6.4 \text{ km}}} \text{ before Q.}$$

$$61.00 = V \cdot 1.2$$

-11-

β) Find trajectory of package.

$$x = 400t \quad ; \quad t = \frac{x}{400}$$

$$y = -5 \left(\frac{x^2}{160,000} \right) + 1280$$

$$y = 1280 - \frac{x^2}{32,000}$$

$$\frac{dy}{dx} = \frac{-2x}{32,000} = \frac{-x}{16,000}$$

$$V = \frac{1280}{6000} \quad \frac{dx}{dt}$$

$$\frac{dx}{dt} = 400$$

$$\dot{x} = V, \quad \dot{y} = -gt$$

$$\text{At } t = 16$$

$$\dot{x} = 400, \quad \dot{y} = -16 \times 10 = -160$$

$$\begin{aligned} \text{Vel. on impact} &= \sqrt{\dot{x}^2 + \dot{y}^2} \\ &= \sqrt{(400)^2 + (-160)^2} \\ &= 430.81 \\ &= 431 \text{ m/s as req'd.} \end{aligned}$$

QUESTION 7

-12-

$$A) \frac{5}{(x-2)(x+3)} = \frac{A}{x-2} + \frac{B}{x+3}$$

$$= \frac{A(x+3) + B(x-2)}{(x-2)(x+3)}$$

$$A(x+3) + B(x-2) = 5$$

$$(A+B)x + 3A - 2B = 5$$

$$A+B=0; A=-B \quad \text{--- (1)}$$

$$3A - 2B = 5. \text{ Sub in (1)}$$

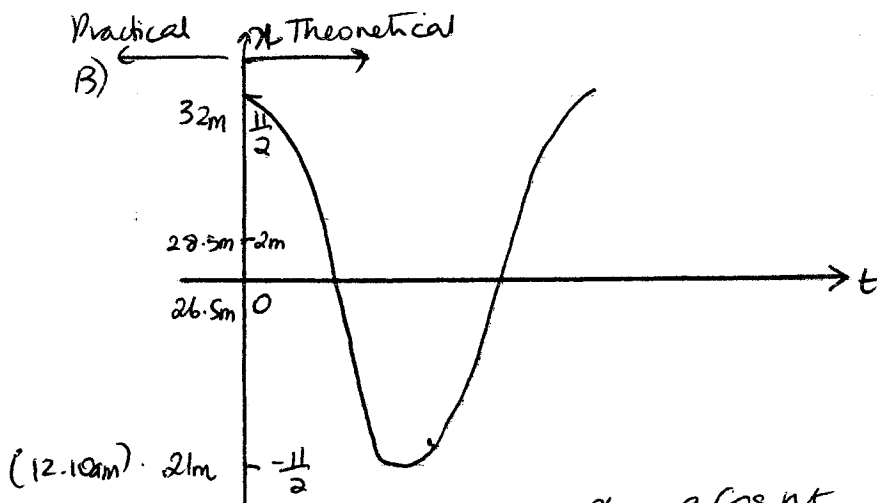
$$3(-B) - 2B = 5; -5B = 5; \underline{B = -1}$$

$$\underline{A = 1}$$

$$\therefore \frac{5}{(x-2)(x+3)} = \frac{1}{x-2} - \frac{1}{x+3}$$

$$ii.) \int \frac{5}{(x-2)(x+3)} dx = \int \frac{1}{x-2} - \frac{1}{x+3} dx$$

$$= \ln(x-2) - \ln(x+3) + C$$



$$\text{Period} = \frac{2\pi}{n} = 13$$

$$n = \frac{2\pi}{13}$$

$$x = a \cos nt$$

$$x = \frac{11}{2} \cos \frac{2\pi}{13} t$$

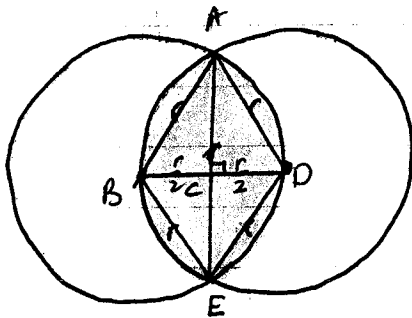
$$2 = \frac{11}{2} \cos \frac{2\pi}{13} t; \cos \frac{2\pi}{13} t = \frac{4}{11}$$

$$\frac{2\pi}{13} t = 1.198$$

$$t = 2 \text{ hrs } 29 \text{ min} \quad \checkmark$$

$\therefore$ earliest time at which ship can enter the harbour =
 $12.10 \text{ am} + 2 \text{ hrs } 29 \text{ min} = \underline{\underline{2.39 \text{ pm}}} \quad \checkmark$

c)



In $\triangle ACD$, $AC^2 = r^2 - \frac{r^2}{4}$ (pythag. theorem)

$$AC^2 = \frac{3r^2}{4}$$

$$AC = \frac{\sqrt{3}r}{2} \quad (AC > 0) \quad \checkmark$$

$$\begin{aligned} \text{Area of } \triangle ACD &= \frac{1}{2} \times \frac{r}{2} \times \frac{\sqrt{3}r}{2} \\ &= \frac{\sqrt{3}r^2}{8} \quad \checkmark \end{aligned}$$

$$\tan \angle AOC = \frac{\sqrt{3}r}{\frac{r}{2}} \times \frac{2}{r}$$

$$\tan \angle AOC = \sqrt{3} \quad ; \quad \angle AOC = \frac{\pi}{3}$$

$$\angle AOC = \frac{\pi}{3}$$

$$\triangle ABC \equiv \triangle ACD \equiv \triangle CDE \equiv \triangle BCE.$$

$$\text{Area of Minor segment} = \frac{\left(\frac{2\pi}{3} \times \pi r^2 \right) - 4 \left(\frac{\sqrt{3}r^2}{8} \right)}{2}$$

$$= \frac{\pi r^2}{6} - \frac{\sqrt{3}r^2}{4} \quad \checkmark$$

$$\text{Area of shaded part} = 4 \left(\frac{\sqrt{3}r^2}{8} \right) + 4 \left(\frac{\pi r^2}{6} - \frac{\sqrt{3}r^2}{4} \right)$$

$$= \frac{\sqrt{3}r^2}{2} + \frac{2\pi r^2}{3} - \sqrt{3}r^2$$

$$= \frac{\sqrt{3}r^2 - 2\sqrt{3}r^2 + 2\pi r^2}{3}$$

$$= r^2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \quad \checkmark$$